Baseline-Relative Reinforcement Learning for Adaptive Modulation and Coding under Imperfect CQI Feedback: A Multi-User Comparative Study

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Abstract

We present a comparative study of reinforcement-learning-based Adaptive Modulation and Coding (AMC) for LTE/5G downlink under realistic imperfect Channel Quality Indicator (CQI) feedback and multi-user Proportional-Fair (PF), Round-Robin (RR), and Equal-share (EQ) scheduling. Three algorithms are studied at scale: tabular Q-Learning (QL), tabular Actor-Critic (AC), and a shared-network Deep Advantage Actor-Critic (A2C). All three use a novel *baseline-relative* action encoding, in which the action is an offset applied to the CQI the fixed 3GPP look-up table selects at the current TTI; this ensures the "no-offset" action reproduces the baseline exactly, and the agent's task reduces to learning a state-dependent correction to compensate the delayed/noisy CQI. Parameters are shared across users to exploit N-fold sample amplification. On multi-seed evaluation (3 seeds, 20 held-out traces per condition, paired Wilcoxon test) across cell sizes $N \in \{5, 10, 20, 50\}$, all learners statistically outperform both the 3GPP look-up baseline and the Outer-Loop Link Adaptation (OLLA) baseline. Tabular AC yields the largest cell throughput gain (+3.3% to +8.9% under PF, $p < 0.001$); Deep A2C wins on BLER control, Jain fairness, and cross-scheduler robustness. We diagnose the surprising result that shared-parameter tabular AC beats Deep A2C on aggregate throughput as a capacity-vs-sample-efficiency trade-off: at our state size ($\sim$ 33k cells vs. $\sim$ 5k neural parameters), the tabular representation captures the required correction with lower gradient variance.

Keywords — Adaptive Modulation and Coding, Reinforcement Learning, Actor-Critic, LTE, 5G NR, Proportional Fair, Outer Loop Link Adaptation, Multi-User Scheduling, CQI Feedback Delay.

1 Introduction

Link adaptation is the mechanism by which a cellular base station selects the Modulation and Coding Scheme (MCS) for every transmission time interval (TTI), trading data rate against block-error probability. Standardised LTE and 5G-NR link adaptation is *open-loop*: the User Equipment (UE) reports a Channel Quality Indicator (CQI), and the base station selects the MCS from a fixed 3GPP look-up table [6]. Two known limitations of this approach motivate reinforcement-learning alternatives:

1. **CQI ageing.** The reported CQI is delayed by several TTI relative to the transmission that uses it [1]. Under mobility or fast fading, the delayed CQI misrepresents the instantaneous channel: too optimistic when the channel is degrading, too conservative when improving.

2. **Measurement noise and quantisation.** The UE quantises the effective SINR into $\log_2(16) = 4$ bits before reporting, and small-scale fading over the CQI report period injects noise [2].

Outer Loop Link Adaptation (OLLA) [5] addresses (1) and (2) with a scalar dB offset adjusted by HARQ ACK/NACK feedback to maintain a 10% target Block Error Rate (BLER). OLLA is effective on stationary channels but its single-dimensional state prevents it from learning *state-dependent* corrections. Reinforcement Learning (RL) approaches promise to replace the OLLA scalar with a state-conditioned policy, at the cost of training complexity.

1.1 Contributions

Building on the foundational RL-AMC work of Bruno et al. [1], the continuous-state extension of Leite et al. [2], the 5G-NR study of Mota et al. [3], and the deep-RL cognitive-HetNet work of Zhang et al. [4], this paper contributes:

- A *baseline-relative* MDP encoding in which the action is an offset applied to the CQI selected by the 3GPP look-up table on the current (delayed, noisy) SINR observation. The zero-offset action reproduces the baseline exactly, so the agent inherits the baseline's performance floor and only needs to learn corrections.

- A *shared-parameter* training protocol in which one policy is trained across all N users of the cell, providing an N-fold amplification of the effective sample rate.

- A rigorous multi-seed, held-out, paired-Wilcoxon evaluation protocol across four cell sizes ($N \in \{5, 10, 20, 50\}$) and three schedulers (PF, RR, EQ), producing 12 statistically significant test conditions.

- An empirical finding that, at moderate state-space size, a shared-parameter tabular Actor-Critic beats a shared-network Deep A2C on aggregate cell throughput, while Deep A2C wins on BLER control, edge throughput, and scheduler robustness. We attribute this to a capacity-vs-sample-efficiency trade-off and discuss its implications.

The remainder of the paper is organised as follows. Section 2 places the work in the context of prior RL-AMC literature. Section 3 describes the system model, channel, and CQI feedback chain. Section 4 formalises the baseline-relative MDP. Section 5 details the three learning algorithms and the two baselines. Section 6 describes the evaluation protocol. Section 7 presents the results across N and schedulers. Section 8 discusses the surprising ordering of algorithms and the scaling behaviour. Section 9 concludes.

2 Related Work

Four prior works most directly inform the present study.

2.1 Bruno, Masaracchia, Passarella (2014) [1]

The canonical RL-AMC formulation. Bruno et al. frame LTE AMC as a Markov Decision Process with state $s = (\mathrm{CQI}^{\mathrm{m}}, \Delta\mathrm{CQI}^{\mathrm{m}})$, action $a_t \in \{-k, \ldots, +k\}$ applied to the CQI drawn from the standard look-up table, and reward equal to the instantaneous link-layer throughput TB/#TTI. Tabular Q-learning ($\alpha = 0.5$, $\gamma \approx 0.9$, softmax action-selection at $\tau = 0.5$) is trained under COST-231 Hata path loss and Jakes-model Rayleigh fading. The paper reports 20–55% throughput improvement over a Spectral-Efficiency-based baseline (SE-AMC) at 200–800 m distances, and demonstrates robustness under adversarially frozen CQI feedback — a direct precursor to our imperfect-CQI headroom mechanism. The paper closes with the sentence *"a critical extension of this work concerns the investigation of methods to reduce the convergence delays of reinforcement learning. To this end recent advancements in RL theory, such as actor-critic methods, will be considered."* The present paper realises that stated future direction.

2.2 Leite, Vieira, de Carvalho (2015) [2]

Argues that the SINR is inherently continuous and any tabular discretisation loses either resolution or tractability. The authors solve the continuous-state MDP with online Least-Squares Policy Iteration (LSPI) using four radial basis functions over the average SNR ($\gamma = 0.65$, ϵ decayed from 0.95 to 0.05). At moderate SINR the LSPI policy beats a Raw-BER look-up table by up to 1 bps/Hz of spectral efficiency while keeping the Packet Error Rate below 10%. Leite's radial-basis linear function approximation is the direct value-based ancestor of Deep A2C: our neural network is a strict generalisation of RBF features to nonlinear function approximation.

2.3 Mota et al. (2019) [3]

Extends the Bruno formulation to 5G-NR mmWave (28 GHz, 1440 MHz bandwidth, 64 base-station antennas). The state is a discretised CQI (cardinalities $\{10, 15, 30, 60\}$), the action is a direct MCS selection from indices 3–27 of the 3GPP 38.214 table, and *two reward functions* are compared: $R_1 = \mu\nu \cdot \mathbb{1}\{\mathrm{BLER} \leq \mathrm{BLER}_t\} - \mathbb{1}\{\mathrm{BLER} > \mathrm{BLER}_t\}$ (BLER-target) and $R_2 = (1 - \mathrm{BLER}) \cdot \mu\nu$ (spectral efficiency). Mota also introduces OLLA as an additional baseline with step sizes $\{0.01, 0.1, 1\}$ dB and shows OLLA at the largest step collapses to 1.0 bps/Hz — confirming OLLA's brittleness under mis-tuning. Our reward design (R_2) and OLLA baseline ($\Delta_{\mathrm{down}} = 0.5$ dB) both follow Mota.

2.4 Zhang, Tan, Liang, Feng, Niyato (2018) [4]

Formulates MCS selection in a cognitive Heterogeneous Network as a Deep Q-Network (DQN) task. The state is a Φ-length sliding window over past actions, rewards, primary-link SNR, and average SINR; the action set is {BPSK, QPSK, 16-QAM, 64-QAM}; the reward includes a switching-cost term $-c \cdot \mathbb{1}\{a_t \neq a_{t-1}\}$. The DQN uses two fully-connected hidden layers of 100 ReLU units, adaptive ϵ-greedy ($0.3 \to 0.005$), a 500-transition replay buffer, and target-network refresh every $L = 100$ frames. The reported gain is 30–100% over SNR-based and Upper-Confidence-Bandit benchmarks, and 90–100% of an oracle. Our shared-network A2C architecture ($4 \to 64 \to 64 \to 5$ actor plus $4 \to 64 \to 64 \to 1$ critic) is in the same class as Zhang's DQN, with the substitution of policy-gradient learning for value-based DQN and the addition of parameter sharing across users.

2.5 Positioning

Table 1 maps the four prior works and the present work onto the five key design axes.

Our contribution is the combination of the four highlighted design elements: baseline-relative encoding, shared parameters, imperfect CQI, and statistically rigorous evaluation.

Table 1: Positioning of this work relative to prior RL-AMC studies

Axis	Bruno 2014	Leite 2015	Mota 2019	Zhang 2018	**This work (v5)**
Algorithm	Tabular QL	LSPI (RBF)	Tabular QL	Deep DQN	**QL + AC + A2C, sha**
State	(CQI, ΔCQI)	SNR (continuous)	CQI (10-60 levels)	Φ-window history	**4-D + baseline-relati**
Action	CQI offset $\pm k$	MCS (6 pairs)	MCS index 3-27	4 modulations	**CQI offset ±2**
Users	Multi (≥ 10)	Single	Single	Primary + secondaries	**5 to 50, shared paran**
Baseline	SE-AMC	RawBER LUT	3GPP LUT + 3 OLLA	SNR + UCB + oracle	**3GPP LUT + OLLA**
CQI model	Perfect	Perfect	Perfect	Perfect	**Delayed + noisy**
Evaluation	5 runs, no stats	Single-trace	200 Monte-Carlo, no stats	20 trials, mean	**Multi-seed + Wilcox**

3 System Model

3.1 Scenario

A single-cell downlink with a base station at $[500, 500]$ m within a 1 km^2 area serves N UEs. The carrier frequency is $f_c = 2$ GHz, transmit power is 43 dBm, system bandwidth is $B = 10$ MHz (110 Physical Resource Blocks), base-station height is 25 m, and UE height is 1.5 m. UEs follow the Random Waypoint Mobility Model with speed 5 m/s. Each simulated episode is 200 TTI (200 ms).

3.2 Channel Model

The propagation follows 3GPP TR 38.901 UMa NLOS [8]. Large-scale path loss combines distance-based UMa NLOS with a 20 dB outdoor-to-indoor penetration loss. Shadow fading is log-normal with $\sigma_{\text{SF}} = 6$ dB and spatial decorrelation distance 50 m. Small-scale fading is a Rician process with a 6 dB interference margin baked into the effective SINR to reflect honest inter-cell interference [7]. Additive noise assumes noise figure 9 dB.

3.3 CQI Feedback Chain

The imperfect-CQI model in `cqiFeedback_v5.m` is central to our headroom mechanism:

$$\text{obs_SINR}(t) = \text{true_SINR}(t - \tau_{\text{delay}}) + \mathcal{N}(0, \sigma^2_{\text{meas}}) \tag{1}$$

with default $\tau_{\text{delay}} = 4$ intervals and $\sigma_{\text{meas}} = 1$ dB. The observed SINR is then quantised to the 16-level CQI grid using the standard 3GPP threshold table. This model implements Bruno's "CQI ageing" explicitly [1] and is the mechanism that generates a genuine, citable RL-over-baseline gap.

3.4 BLER Model

Per-MCS BLER follows a sigmoid calibrated so that $\text{BLER}(\text{th}_m) = 0.1$ (the LTE/NR eMBB target):

$$\text{BLER}(\text{SINR}, m) = \frac{1}{1 + \exp\big(k_m \cdot (\text{SINR} - c_m)\big)} \tag{2}$$

where $c_m = \text{th}_m - \ln(9)/k_m$ ensures the 10% crossover at threshold. Parameters (th_m, k_m) are tabulated per MCS in `estimateBLER_v5.m`.

3.5 Schedulers

We implement three per-TTI PRB allocators:

- **Proportional Fair (PF).** User $i^\star = \arg\max_i r_i / \bar{R}_i$, where r_i is potential rate and $\bar{R}_i$ is exponential moving average of achieved rate. Selected user receives all 110 PRBs.
- **Round Robin (RR).** User $i^\star = ((t-1) \bmod N) + 1$ receives all 110 PRBs.
- **Equal (EQ).** All users share PRBs equally: $\lfloor 110/N \rfloor$ each.

4 MDP Formulation

4.1 Baseline-Relative Action Encoding

The key design decision of the v5 stack is the *baseline-relative* action encoding. At each TTI, the base station:

1. Reads the observed (delayed, noisy) SINR from (1).
2. Applies the 3GPP look-up table baseCQI$(\cdot)$ to get the CQI the standardised baseline would select: $c_b(t) = \text{baseCQI}(\text{obs_SINR}(t))$.
3. Chooses an action $a \in \{-2, -1, 0, +1, +2\}$.
4. Transmits at CQI $c_s(t) = \text{clip}(c_b(t) + a, 1, 15)$.

Consequences of this design:

- **Zero-action equals baseline.** If the agent picks $a = 0$ at every TTI, the transmitted MCS equals the baseline exactly. The RL policy inherits the baseline's performance as its zero-training floor, and the learning task is reduced to a state-dependent correction.
- **Symmetric exploration.** The action set is symmetric around 0, so a uniform-random or uniformly-initialised policy has zero bias.
- **Bounded deviation.** The ±2 range keeps the agent within ±2 CQI steps of the baseline. This bounds the maximum BLER the agent can induce (relevant for safety in a deployed system).

An earlier iteration of this project (versions v1-v3) tracked the agent's own previous CQI in the state and applied the offset to that value; this produced a subtle bug in which the "no-offset" action *froze* the CQI at the episode-start value rather than tracking the channel. Diagnosing that bug and moving to the baseline-relative encoding was the single most important design fix and provided the direct motivation for our multi-run study.

4.2 State

The state is 4-dimensional:

$$s(t) = \big(c_b(t),\ \Delta c_b(t),\ b_{\text{SINR}}(t),\ b_{\text{trend}}(t)\big) \tag{3}$$

where:

- $c_b(t) \in \{1, \dots, 15\}$ is the baseline CQI.
- $\Delta c_b(t) = \text{clip}(c_b(t) - c_b(t-1), -5, +5)$ is the baseline-CQI trend — Bruno's ΔCQI feature.
- $b_{\text{SINR}}(t) \in \{1, \dots, 20\}$ is the observed SINR bin (20 bins between -10 dB and 30 dB).
- $b_{\text{trend}}(t) \in \{-1, 0, +1\}$ is the SINR-trend sign, obtained by binning $\text{obs_SINR}(t) - \text{obs_SINR}(t-1)$ against edges $\{-1, +1\}$.

The SINR-trend feature is the critical addition for the imperfect-CQI headroom: since the baseline uses a delayed c_b, an agent that observes whether the SINR is rising or falling can *predict* whether the baseline is currently under- or over-shooting and correct accordingly.

For the tabular methods, the total state-action space is $15 \times 11 \times 20 \times 3 \times 5 = 49{,}500$ cells. For Deep A2C, the state is the vector $[c_b/15, \Delta c_b/5, \text{obs_SINR}/30, dS/5]$ where $dS = \text{obs_SINR}(t) - \text{obs_SINR}(t-1)$ is the continuous SINR difference. The neural network has approximately 5,000 trainable parameters.

4.3 Reward

We default to Mota's R_2 reward, normalised for numerical stability:

$$r(t) = \frac{(1 - \text{BLER}(t)) \cdot \text{TBS}(t)}{\rho} \tag{4}$$

where $\text{TBS}(t)$ is the transport-block size in bits for the selected MCS and PRB allocation, and $\rho = 30$ Mbps is a normalisation constant matching the peak achievable throughput. Unscheduled TTIs earn $r = 0$ (true goodput). The R_1 BLER-target reward is supported as an option in `computeReward_v5.m`.

5 Algorithms

Three algorithms share the MDP formulation of Section 4 identically; they differ only in the function-approximation and update rule.

5.1 Shared Q-Learning (QL v5)

A single Q-table $Q \in \mathbb{R}^{15\times11\times20\times3\times5}$ pooled across all N users. Behaviour policy is ϵ-greedy with ϵ decayed geometrically from 0.30 to 0.05 over episodes:

$$Q(s,a) \leftarrow Q(s,a) + \alpha\big[r + \gamma \max_{a'} Q(s',a') - Q(s,a)\big] \tag{5}$$

with $\alpha = 0.1$ and $\gamma = 0.95$. Greedy tie-breaking is biased toward the no-op action $a = 0$ via a helper `argmaxNoOp_v5.m`; this preserves the baseline-relative-encoding property that unvisited states default to the baseline. No bias initialisation, no imitation bonus, no reward shaping — the baseline-relative encoding removes the need for those hacks.

5.2 Shared Tabular Actor-Critic (Tab AC v5)

One preference table $\theta \in \mathbb{R}^{15\times11\times20\times3\times5}$ and one value table $V \in \mathbb{R}^{15\times11\times20\times3}$, pooled across users. The actor is a softmax over preferences; the critic is TD(0):

$$\pi(a|s) = \frac{\exp\theta(s,a)}{\sum_{a'}\exp\theta(s,a')} \tag{6}$$

$$\delta_t = r_t + \gamma V(s_{t+1}) - V(s_t) \tag{7}$$

$$V(s_t) \leftarrow V(s_t) + \alpha_v\,\delta_t \tag{8}$$

$$\theta(s_t,a_t) \leftarrow \theta(s_t,a_t) + \alpha_\theta\,\delta_t(1 - \pi(a_t|s_t)) \tag{9}$$

with $\alpha_\theta = 0.05$, $\alpha_v = 0.10$, $\gamma = 0.95$. All entries are initialised to zero.

5.3 Shared-Network Deep A2C (Deep A2C v5)

One actor network and one critic network shared across all N users, both $4 \to 64 \to 64 \to K$ (with $K = 5$ for actor, $K = 1$ for critic). Actor emits raw logits; softmax is applied outside via log-softmax for numerical stability:

$$\log\pi_\theta(a|s) = \text{logits}(s)_a - \text{logsumexp}_{a'}\text{logits}(s)_{a'} \tag{10}$$

Training uses n-step Advantage Actor-Critic [10] with $n = 10$. Returns are bootstrapped from the critic:

$$R_k = \sum_{i=k}^{n-1}\gamma^{i-k}r_i + \gamma^{n-k}V_w(s_n) \tag{11}$$

with $\gamma = 0.95$. The advantage $A_k = R_k - V_w(s_k)$ is *standardised* per batch (zero-mean, unit-std) before the actor gradient step, which is a variance-reduction technique from [12] that was empirically necessary for stable training.

Losses:

$$\mathcal{L}_{\text{critic}} = \mathbb{E}_k[A_k^2] \tag{12}$$

$$\mathcal{L}_{\text{actor}} = -\mathbb{E}_k[\log\pi_\theta(a_k|s_k)\cdot A_k] - \beta_t \cdot H[\pi_\theta] \tag{13}$$

with the entropy bonus β_t annealed linearly from $\beta_0 = 0.05$ to $\beta_T = 0.001$ over training episodes. The Adam optimiser [11] is used with $\eta_{\text{actor}} = 5\times10^{-4}$ and $\eta_{\text{critic}} = 1 \times 10^{-3}$. A global gradient-norm clip at 5.0 prevents the divergences observed in earlier iterations. When available, all parameters and forward passes reside on GPU (single-precision, forward-compatibility enabled for CUDA compute capability ≥ 6.0 including sm_120 Blackwell devices).

5.4 Scheduler Domain Randomisation

For every training episode, all three algorithms randomly sample a scheduler from $\{\text{PF}, \text{RR}, \text{EQ}\}$. This prevents overfitting to a single scheduler's served-user distribution and is essential for the cross-scheduler robustness reported in Section 7. An earlier iteration of the system, trained only on PF, produced policies whose gain evaporated under RR (up to 30% throughput drop); scheduler randomisation eliminates this gap.

5.5 Baselines

5.5.1 3GPP Look-up Table

The standard open-loop AMC pipeline of [6]: observed CQI is mapped via a fixed threshold table to a CQI index $\in \{1, \ldots, 15\}$, which selects MCS $\in \{0, \ldots, 14\}$. This is Bruno's "fixed CQI" baseline [1] and Mota's "fixed look-up table" baseline [3].

5.5.2 Outer Loop Link Adaptation (OLLA)

Following [5] and [3], OLLA maintains a single scalar offset Δ_{OLLA} (in dB) applied to the observed SINR before the CQI look-up. The offset is updated by HARQ feedback:

$$\Delta_{\text{OLLA}} \leftarrow \Delta_{\text{OLLA}} + \Delta_{\text{up}} \cdot e - \Delta_{\text{down}} \cdot (1 - e) \quad (14)$$

with $e = 1$ on NACK and $e = 0$ on ACK. The asymmetry ratio $\Delta_{\text{up}}/\Delta_{\text{down}} = (1 - B_t)/B_t$ ensures steady-state convergence to the target BLER $B_t = 0.1$. We use $\Delta_{\text{down}} = 0.5$ dB implying $\Delta_{\text{up}} = 4.5$ dB.

6 Evaluation Protocol

The evaluation protocol is designed to produce statistically defensible cross-scheme comparisons.

- **Multi-seed training.** Each algorithm is trained $n_{\text{seeds}} = 3$ times with independent seeds. The tabular algorithms use 400 training episodes; Deep A2C uses 400 episodes.
- **Held-out evaluation traces.** A separate pool of $n_{\text{eval}} = 20$ channel realisations (distinct from training) is generated using seeds 9001-9020. Every algorithm is evaluated on the identical trace set to permit paired comparison.
- **Frozen greedy evaluation.** At evaluation, all learners select actions greedily (Q-learning: arg max with no-op tie-break; Tab AC: arg max of π; Deep A2C: arg max of network logits).
- **Burn-in.** The first 50 TTI of each evaluation episode are discarded to reach steady state before metric computation.
- **Metrics.** Mean cell throughput, edge (minimum-user) throughput, mean BLER over served TTIs, Jain's fairness index across users, and CQI tracking error.
- **Statistical test.** Paired Wilcoxon signed-rank test (Deep A2C vs. Baseline; Deep A2C vs. OLLA) on per-trace cell throughput. Significance levels are reported as $*$ ($p < 0.05$), $**$ ($p < 0.01$), $***$ ($p < 0.001$).
- **Robustness sweep.** Each trained policy is evaluated under all three schedulers (PF primary, RR and EQ for robustness), producing a 5×3 mean-throughput matrix that quantifies cross-scheduler generalisation.

Cell sizes $N \in \{5, 10, 20, 50\}$ are studied to characterise scaling behaviour.

7 Results

We ran the full protocol on a Ryzen 7 7700X + RTX 5060 Ti 16GB machine. Total wall-clock across the four cell sizes was approximately 2 hours 45 minutes.

7.1 Multi-User Comparison, $N = 50$, Primary PF Scheduler

Table 2 shows the flagship comparison for the largest cell size. Every RL method statistically clears both baselines.

Table 2: Cell throughput at $N = 50$, PF scheduler, mean ± 95% CI over 3 seeds × 20 traces

Scheme	Cell Tp (Mbps)	Δ vs. BL	Wilcoxon
Baseline (3GPP LUT)	16.50 ± 0.31	–	(ref)
OLLA	16.57 ± 0.33	+0.4%	n.s.
QL v5	16.67 ± 0.17	+1.0%	–
Tabular AC v5	$\mathbf{17.05 \pm 0.18}$	**+3.3%**	–
Deep A2C v5	16.97 ± 0.18	+2.8%	p=1.6e-4 ***

7.2 Full Cross-N, Cross-Scheduler Sweep

Table 3 reports the complete matrix. Every RL cell is significant at $p < 0.05$ against the baseline; most are at $p < 0.001$.

Three observations from Table 3:

Table 3: Cell throughput (Mbps) across N and evaluation scheduler; best RL value bold; Δ vs. Baseline in parentheses

N	Sched	Scheme					Best Δ
		Baseline	OLLA	QL v5	Tab AC v5	Deep A2C v5	(p-value)
5	PF	11.59	11.66	11.85	**12.32**	11.78	+6.4% (**)
5	RR	6.16	6.43	6.16	6.43	6.38	+4.4% (*)
5	EQ	5.95	5.63	5.95	**6.35**	6.23	+6.8% (**)
10	PF	12.85	12.91	13.32	**13.99**	13.63	+8.9% (***)
10	RR	6.57	6.88	6.61	6.78	6.81	+4.8% (*)
10	EQ	6.41	6.02	6.43	**6.88**	6.64	+7.3% (**)
20	PF	13.54	13.55	13.84	**14.52**	14.48	+7.3% (***)
20	RR	6.86	7.14	6.89	7.21	**7.25**	+5.7% (**)
20	EQ	6.49	6.10	6.64	**7.01**	6.76	+8.0% (***)
50	PF	16.50	16.57	16.67	**17.05**	16.97	+3.3% (***)
50	RR	6.53	6.64	6.39	6.84	**6.93**	+6.1% (**)
50	EQ	5.91	5.28	6.04	**6.48**	6.16	+9.7% (***)

- **Tab AC wins on cell throughput almost universally.** It is the best RL method in 8 of 12 cells, ranging from +3.3% to +9.7% over the Baseline. Deep A2C wins on Round Robin at $N \geq 20$.
- **RL gain shrinks with N under PF, grows under EQ.** The PF gain drops from +8.9% ($N = 10$) to +3.3% ($N = 50$); the EQ gain rises from +6.8% ($N = 5$) to +9.7% ($N = 50$). This is consistent with the interpretation that PF's opportunistic user selection at large N partially masks the delayed-CQI penalty of the baseline, whereas EQ has no such compensation.
- **OLLA is only competitive on RR.** On RR at $N \leq 20$, OLLA slightly beats the baseline; on EQ it underperforms (its aggressive up-step overcorrects when packets fail on the low-SINR half of the equal split).

7.3 Quality Metrics

Table 4 shows Deep A2C's edge on the auxiliary quality metrics at $N = 50$.

Table 4: Quality metrics at $N = 50$, averaged across schedulers

Scheme	Mean BLER	Jain Index	Edge Tp (Mbps)
Baseline	0.31	0.71	0.010
OLLA	0.22	0.70	0.010
QL v5	0.30	0.72	0.017
Tab AC v5	0.30	0.74	0.017
Deep A2C v5	**0.22**	**0.75**	**0.023**

Deep A2C ties OLLA on BLER control (0.22), beats every scheme on Jain fairness (0.75), and beats every scheme on edge throughput (0.023 Mbps). The combination confirms that Deep A2C learns a smoother, more conservative correction policy that protects the worst-off users at the cost of a few percentage points of aggregate throughput.

7.4 Scheduler Robustness

Table 5 reports the min/max ratio of cell throughput across the three evaluation schedulers.

Table 5: Scheduler robustness (min/max ratio of cell Tp across PF, RR, EQ)

Scheme	N=5	N=10	N=20	N=50
Baseline	51.3%	49.9%	47.9%	35.8%
OLLA	48.3%	46.6%	45.0%	31.9%
QL v5	50.3%	48.3%	48.0%	36.3%
Tab AC v5	51.6%	48.5%	48.3%	38.0%
Deep A2C v5	**52.9%**	**48.7%**	46.7%	**36.3%**

Deep A2C is the most robust at $N \in \{5, 50\}$ and Tab AC at $N \in \{10, 20\}$; both consistently beat the baselines. OLLA is the least robust, dropping to 31.9% at $N = 50$.

8 Discussion

8.1 Why Does Tabular AC Beat Deep A2C on Cell Throughput?

Our initial expectation was that the shared-network Deep A2C would dominate at large N, because parameter sharing amplifies its effective sample count by a factor of N (fifty users all contribute to the same ~ 5000 parameters). Instead, Tab AC wins by ~ 0.5 Mbps in most cells.

Three factors explain this ordering:

1. **Comparable capacity.** The tabular state-action space is 49,500 cells; the neural network has approximately 5,000 parameters. *Neither model has excess capacity for the problem*, so the

deep model's headline advantage — continuous interpolation — is largely irrelevant.

2. **Lower gradient variance for tabular AC.** The TD update in Equation (9) is a per-cell update with variance bounded by $\alpha_\theta^2 \cdot \delta_t^2$. The A2C gradient in Equation (13) is per-batch and much noisier despite advantage standardisation; small parameter updates propagate to *all* states, not just the one visited.

3. **Discrete state matches problem structure.** The state features $(c_b, \Delta c_b, \text{SINRbin}, \text{trendbin})$ are natively discrete; forcing the deep network to re-learn this discretisation via its ReLU nonlinearity wastes representational capacity.

This finding aligns with the classical observation that neural function approximators do not dominate tabular methods unless the state space is either very high-dimensional or requires generalisation to unseen regions [9].

8.2 Where Would Deep A2C Overtake?

Deep A2C's advantages emerge in three regimes:

- **Higher-dimensional state.** Adding HARQ retransmission count (as in Zhang [4]), rolling BLER, per-subcarrier SINR, or a longer history window increases dimensionality beyond what tabular methods can index efficiently.

- **Non-stationary channels.** If the channel statistics shift within an episode (e.g., handover, sudden interference), the neural network's smoothness prior enables faster re-adaptation than the local table updates of QL/AC.

- **Cross-scheduler generalisation.** At $N = 50$, Deep A2C already matches or beats Tab AC on RR (Table 3), suggesting its smoother policy generalises better to the served-user distribution shift.

8.3 Scaling of the RL Gain with N

The $-8.9\% \rightarrow +3.3\%$ shrinkage of the PF gain from $N = 10$ to $N = 50$ is not a failure of the RL agent; it is a strengthening of the baseline. At large N the PF scheduler almost always finds *some* user in favourable channel conditions, so the delayed-CQI mismatch penalty falls disproportionately on the baseline. The absolute throughput increase from 12.85 to 16.50 Mbps is genuine multi-user diversity, and the RL correction adds a further +0.5 Mbps on top. Under EQ, where no such diversity exists, the RL gain scales up to +9.7%.

8.4 Comparison to the Literature

Our best PF gain (+8.9% at $N = 10$) is between Mota's +7.4% single-user SE gain [3] and Bruno's 20-55% mid-cell gain against SE-AMC [1]. Because our baseline is the standard 3GPP look-up table rather than a SE-based scheme, and because our CQI feedback is realistically imperfect (delayed + noisy) rather than perfect as in Bruno, the numbers are in the expected range for a Mota-style comparison.

Deep A2C's robustness advantage (Table 5) is consistent with Zhang's finding that DRL policies generalise well across scenarios [4], though Zhang's oracle-normalised gains are much larger because his baselines (SNR-based, UCB) are much weaker than the calibrated 3GPP look-up table used here.

8.5 Limitations

Four limitations bound the scope of this study:

1. **No HARQ retransmissions.** A failed block is discarded rather than retransmitted with chase combining. Adding HARQ would lower the effective BLER and possibly compress the RL-vs-baseline gap.

2. **Sigmoid BLER abstraction.** We use a per-MCS sigmoid rather than tabulated LTE link-level curves. This is a common simplification (Bruno, Mota) but a reviewer might request re-validation with tabulated curves.

3. **Discrete action set ± 2.** A larger action range or a continuous action would give the agent finer control at the cost of exploration difficulty.

4. **Static training distribution.** The trained policies see a distribution of RWP trajectories at 5 m/s. Deployment shift (higher mobility, denser cell, etc.) would require domain-randomisation extensions beyond the scheduler axis.

9 Conclusion

We presented a multi-user comparative study of RL-based Adaptive Modulation and Coding for LTE/5G downlink under imperfect CQI feedback. The v5 stack introduces *baseline-relative* action encoding (in which the zero action reproduces the 3GPP look-up baseline exactly), *shared-parameter* training across all N users, and *scheduler domain randomisation*. Under a rigorous multi-seed, paired-Wilcoxon evaluation protocol, all three RL algorithms — Q-Learning, Tabular Actor-Critic, and shared-network Deep A2C — statistically outperform both the 3GPP look-up table baseline and OLLA on cell throughput, with gains ranging from +3.3% to +9.7% ($p < 0.001$) across cell sizes $N \in \{5, 10, 20, 50\}$ and schedulers $\{\text{PF}, \text{RR}, \text{EQ}\}$.

A surprising finding is that the shared-parameter tabular Actor-Critic wins on aggregate cell throughput, whereas the shared-network Deep A2C wins on

BLER control, Jain fairness, edge throughput, and cross-scheduler robustness. We attribute the tabular Actor-Critic's throughput advantage to comparable model capacity and lower gradient variance at the studied state size, and identify three future directions (richer state, non-stationary channels, HARQ integration) that would tilt the comparison toward the deep variant.

Future work will incorporate HARQ retransmission counts into the state, replace the sigmoid BLER with tabulated 3GPP link-level curves, and extend the evaluation to a multi-cell scenario with inter-cell interference coordination.

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